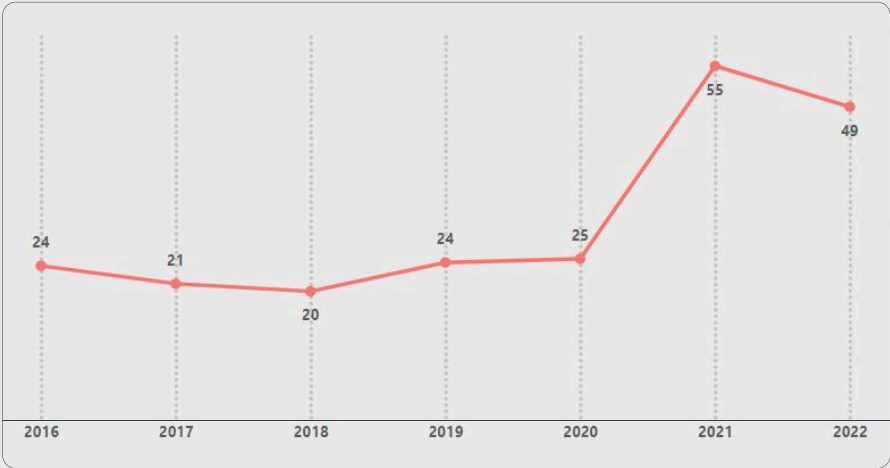


The High Risk of Misusing and Overusing

ANTIBIOTICS: A Threat to the Health of Bangladeshis

Antibiotics misuse is a pressing public health challenge in Bangladesh, contributing to the rise of antimicrobial resistance (AMR) and undermining the efficacy of these life-saving drugs. With a densely populated country and a healthcare system struggling with accessibility and regulation, the misuse of antibiotics has become widespread across various sectors of society.

According to **Global Antimicrobial Resistance and Use Surveillance System (GLASS)** data, Bangladesh ranked as the **3rd** largest consumer of antibiotics globally in 2022.



This graph highlights antibiotic usage trends (DDD per 1,000 inhabitants) from 2016 to 2022. A sharp spike occurred in Bangladesh during the **COVID-19 pandemic**, likely due to concerns over secondary infections and increased antibiotic prescriptions.

Patterns of Antibiotic Usage in Bangladesh

Antibiotics misuse is a pressing public health challenge in Bangladesh, contributing to the rise of antimicrobial resistance (AMR) and undermining the efficacy of these life-saving drugs. With a densely populated country and a healthcare system struggling with accessibility and regulation, the misuse of antibiotics has become widespread across various sectors of society.

Antibiotic use in Bangladesh spans across healthcare facilities, self-medication practices, and agricultural sectors. Some key patterns include:

- 1. Over-the-Counter Availability:** In Bangladesh, antibiotics are often sold without a prescription, making them readily accessible. Many individuals self-diagnose and purchase antibiotics from pharmacies or local drug stores, often without proper guidance.
- 2. Inappropriate Prescriptions:** A significant portion of antibiotic prescriptions in Bangladesh are either unnecessary or inappropriate. Physicians often prescribe broad-spectrum antibiotics for viral infections, such as the common cold or flu, where antibiotics are ineffective.
- 3. Self-Medication and Dosage Non-Compliance:** Many people discontinue antibiotics as soon as they feel better, failing to complete the prescribed course. This non-compliance accelerates the development of resistant bacteria.
- 4. Poor Healthcare Infrastructure:** In rural areas, where access to qualified healthcare providers is limited, antibiotics are often prescribed by untrained or informal healthcare practitioners, leading to misuse and overuse.



Antibiotic usage in Bangladesh by route and AWaRe classification (2022)

Oral formulations of antibiotics are the preferred formulations used in primary care and account for over 90% of the total use in many countries. Use of oral formulations is a surrogate for consumption in the community (i.e., outside hospitals). Most non-severe infections can be safely treated with oral antibiotics and this approach is encouraged as it has several advantages, for example avoidance of risks associated with vascular access lines (infections, thrombosis) and avoidance of hospitalization. Countries with a high relative use of parenteral antibiotics (e.g., >15-20%) should investigate the underlying reasons since it may reflect inappropriate use of parenteral antibiotics which could be a target for antibiotic stewardship interventions.

In Bangladesh, **97%** of antibiotic usage is administered **orally**.

Watch antibiotics such as ciprofloxacin and ceftriaxone are broader spectrum antibiotics, generally with higher price and are recommended in the WHO AWaRe antibiotic book only as first-choice options for patients with more severe clinical presentations or for infections where the causative pathogens are more likely to be resistant to Access antibiotics, such as upper urinary tract infections (UTIs). Watch antibiotics are key targets for antibiotic stewardship interventions.

The **Watch** category, recommended for more **severe clinical infections**, is the **most used** in Bangladesh and the **most expensive** in the entire world.

Factors Driving Antibiotics Misuse

Several socio-economic and systemic factors contribute to the misuse of antibiotics in Bangladesh:

- **Lack of Awareness:** A large portion of the population is unaware of the dangers of improper antibiotic use.
- **Weak Regulatory Mechanisms:** Although regulations exist to control the sale of antibiotics, enforcement is often lax.
- **Inadequate Training:** Many healthcare providers lack updated knowledge about antibiotic stewardship.
- **Cultural Practices:** There is a cultural tendency to demand antibiotics from healthcare providers for even minor illnesses.



Strategies to Combat Antibiotic Misuse in Bangladesh

Tackling antibiotics misuse in Bangladesh requires a multi-faceted approach:

- **Strengthening Regulations:** Strict enforcement of prescription-only antibiotic sales can limit over-the-counter availability.
- **Awareness Campaigns:** Public education initiatives can inform people about the dangers of misuse and the importance of completing prescribed courses.
- **Capacity Building in Healthcare:** Training programs for healthcare providers, including informal practitioners, can promote rational antibiotic prescribing.
- **Improving Healthcare Access:** Enhancing the accessibility and affordability of healthcare services can reduce dependence on untrained practitioners.
- **Antimicrobial Stewardship Programs:** Hospitals and clinics should adopt stewardship programs to monitor and optimize antibiotic use.
- **Research and Surveillance:** Continuous monitoring of antibiotic resistance patterns can guide effective policy-making and intervention.

Antibiotic misuse in Bangladesh is a multifaceted issue requiring urgent attention. Without immediate and sustained action, the country risks facing a public health catastrophe fueled by antimicrobial resistance. Collaborative efforts involving government bodies, healthcare providers, the agricultural sector, and civil society are essential to mitigate this crisis and safeguard the efficacy of antibiotics for future generations.